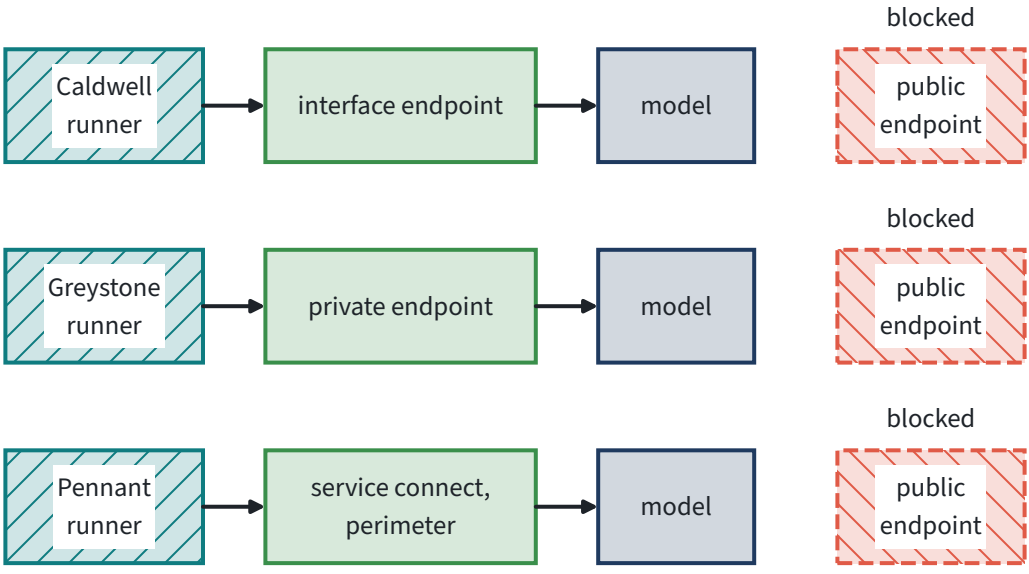


checks N1 to N3 prove both directions